

Black Box Testing Techniques: Amazon Application Audit

Target Feature: Quantity Selector & Order Lifecycle

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1. Equivalence Partitioning (EP)

Feature under test: Quantity Selector Input Field (Accepts values 1 to 30)

- **Partition 1: Invalid-Low**
 - *Range:* All integers ≤ 1
 - *Test Value:* 0
 - *Expected Result:* System displays a validation error or automatically defaults the field to 1.
- **Partition 2: Valid**
 - *Range:* All integers from 1 to 30
 - *Test Value:* 15
 - *Expected Result:* System successfully accepts the entry and updates the cart quantity to 15.
- **Partition 3: Invalid-High**
 - *Range:* All integers ≥ 31
 - *Test Value:* 35
 - *Expected Result:* System displays a validation error or caps the selection at a maximum of 30.

2. Boundary Value Analysis (BVA)

Feature under test: Quantity Selector Input Field Boundary Limits (1 and 30)

- **Test Value: 0** (Just below the lower valid boundary)
 - *Expected Result:* System rejects the input or defaults to 1.
- **Test Value: 1** (The exact lower boundary limit)
 - *Expected Result:* System successfully accepts a quantity of 1.
- **Test Value: 2** (Just above the lower valid boundary)
 - *Expected Result:* System successfully accepts a quantity of 2.
- **Test Value: 29** (Just below the upper valid boundary)
 - *Expected Result:* System successfully accepts a quantity of 29.
- **Test Value: 30** (The exact upper boundary limit)

- *Expected Result:* System successfully accepts a quantity of 30.
- **Test Value:** 31 (Just above the upper valid boundary)
 - *Expected Result:* System rejects the input or caps the selection at 30.

3. Decision Table Testing — Amazon Free Shipping Rules

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
Is Prime Member?	Yes	No	No	No	No
Order Total > \$35?	Don't Care	Yes	Yes	No	Don't Care
Items Eligible?	Yes	Yes	No	Yes	No
Free Shipping Applied?	YES	YES	NO	NO	NO

4. State Transition Testing — Order Lifecycle Workflow

Tracked Order Pipeline : Placed —> Shipped —> Out for Delivery —> Delivered —> Returned

- **Transition 1: Placed —> Shipped**
 - *Trigger Action:* Warehouse packs items and handovers the package to the carrier.
 - *Expected Result:* Order status changes from "Placed" to "Shipped"; tracking number is generated.
- **Transition 2: Shipped —> Out for Delivery**
 - *Trigger Action:* Local delivery hub scans package onto a delivery truck.
 - *Expected Result:* Order status changes to "Out for Delivery"; estimated delivery window updates to "Today".

- **Transition 3: Out for Delivery —> Delivered**
 - *Trigger Action:* Courier drops off package and marks it as delivered in their handheld device.
 - *Expected Result:* Order status updates to "Delivered"; confirmation email/notification is sent to user.
- **Transition 4: Delivered —> Returned**
 - *Trigger Action:* Customer drops package off at a return drop point, and the return barcode is scanned.
 - *Expected Result:* Order status updates to "Returned"; refund process is automatically triggered.